

MD RAKIBUL ISLAM RAIHAN (杨瀚)

AI Engineer | LLM Inference & GPU Systems | Multimodal AI Research — New Graduate 2027 (China / Hong Kong)

Email: raihanrakib.143@gmail.com | Phone: +86-17722362949 | Xi'an, China

LinkedIn: linkedin.com/in/theraihanrakib | GitHub: github.com/theraihanrakibb

Portfolio: theraihanrakib.github.io/Online-Portfolio/

Languages: English (Fluent) | Chinese (Conversational) | Bangladesh

SUMMARY

M.Eng. Software Engineering student (**Top 1%**, NWPU) with hands-on AI engineering across LLM inference systems, GPU optimization, and multimodal AI research. **First author** of an IEEE TNNLS (Q1) paper (NVFP4-DiT) — a **4-bit audio-guided video diffusion transformer** for low-cost AIGC video generation (**4× memory reduction**). AI Infrastructure Engineer Intern at InfiX.ai: built production LLM serving on **4× A800** clusters (**92.27% cache hit**, **8.3× TTFT**, **192→256** concurrency, **FP8 quantization**). Research spans video/image generation, multimodal deepfake detection, and low-precision (FP8/FP4/NVFP4) inference.

EXPERIENCE

AI Infrastructure Engineer Intern | InfiX.ai | Shenzhen, China | Apr 2026 – Jun 2026

- Patched SGLang source code across cache management and memory allocation modules (kvcacheio.py, hiradix_cache.py, memory_pool_host.py) to resolve production cache failures and support deployment of hybrid-attention Qwen models on **4× A800** GPU clusters.
- Designed a 3-tier KV-cache architecture (GPU HBM → CPU DRAM → SSD) using SGLang and Mooncake RDMA, achieving **92.27% cache hit rate** and reducing TTFT by **8.3× (55s → 6.7s)**.
- Profiled CUDA/Triton inference pipelines with Nsight Systems, identifying GEMM bottlenecks and improving serving efficiency through kernel optimization and **FP8 quantization**.
- Built PD-disaggregated serving architecture with RDMA-based KV sharing, increasing stable concurrency from **192 to 256** requests.
- Diagnosed and resolved **12+** production issues involving RDMA, etcd, networking, storage, and distributed inference systems.

Software Engineer Intern (AI Agent) | Hong Kong Amram Group | Shenzhen, China | Jun 2025 – Sep 2025

- Developed production full-stack applications using React, REST APIs, and backend automation services.
- Built AI-agent workflow automation systems that improved operational efficiency by **25%**.

Electrical Software Engineer Intern | Shaanxi Longong Intelligent Technology | Xi'an, China | Feb 2025 – Apr 2025

- Developed PLC-based industrial control systems using CODESYS for robotics and industrial automation.
- Implemented Modbus, CAN, and Ethernet/IP communication protocols, improving motor-control performance by **15%**.

RESEARCH

Graduate Researcher | Northwestern Polytechnical University | Sep 2024 – Present

- Developed **NVFP4-DiT**, a **4-bit audio-guided video diffusion transformer** achieving **4× memory reduction** while maintaining generation quality.
- Implemented low-precision quantization (**FP8/FP4/NVFP4**) for multimodal diffusion transformers; evaluated quality-efficiency trade-offs.
- First author, "NVFP4-DiT: Efficient 4-Bit Audio-Guided Video Diffusion Transformers for Low-Cost Video Generation" (IEEE Transactions on Multimedia, under review).
- Benchmarked NVFP4-DiT on **A800/H100 GPUs**; profiled end-to-end latency and memory with PyTorch Profiler and Nsight to guide low-precision kernel choices.
- Research on multimodal deepfake detection using visual, audio, and temporal consistency analysis.
- M.Eng. Thesis: "Detecting Deepfake Videos: A Multimodal AI Framework Leveraging Visual, Auditory, and Temporal Inconsistencies."

Undergraduate Researcher | Northwestern Polytechnical University | Jan 2023 – Jul 2024

- Designed a distributed confidential-query framework using Apache Spark and CKKS homomorphic encryption.
- Achieved **95% privacy preservation** and **20% performance improvement** via optimized analytics pipelines.
- B.Eng. Thesis: "Design and Implementation of a Distributed Confidential Query Protocol for Apache Spark."

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, Bash, CUDA C++

LLM Inference & Serving: SGLang (RadixAttention, paged attention, continuous batching), vLLM, TensorRT-LLM, KV-cache design, PD disaggregation, prefix caching, Mooncake RDMA, NCCL, etcd, NATS

GPU & Performance: CUDA, CUTLASS, Triton, FlashAttention-2/3, FP8/FP4/INT4 quantization, GEMM optimization, Nsight Systems/Compute, PyTorch Profiler

ML & Distributed Training: PyTorch, DeepSpeed, Megatron-LM, LoRA/QLoRA, ZeRO, distributed training/inference, Apache Spark

Generative Vision & Multimodal: Diffusion (DDPM, DiT), audio-guided video generation, AIGC video/image, multimodal understanding & fusion, deepfake detection, OpenCV

Infrastructure / Familiar: Linux, Docker, Kubernetes, Slurm, Prometheus, Grafana, TCP/IP, Git, CI/CD, AWS / Aliyun cloud, ML compilers (XLA/TVM), TensorRT

EDUCATION

M.Eng. Software Engineering, Northwestern Polytechnical University (985/211) — Sep 2024 – Mar 2027

School of Software · GPA **88/100** · **Top 1%** · Chinese Government Scholarship

B.Eng. Computer Science & Technology, Northwestern Polytechnical University (985/211) — Sep 2020 – Jul 2024

School of Computer Science · GPA **85/100** · **Top 1%** · Presidential Scholarship · Best Undergraduate Thesis Award

PUBLICATIONS & HONORS

Publication: First Author, "NVFP4-DiT: Efficient 4-Bit Audio-Guided Video Diffusion Transformers for Low-Cost Video Generation" — IEEE Transactions on Neural Networks and Learning Systems (TNNLS, Q1), Under Review.

Honors: Chinese Government Scholarship (2024–2026); Presidential Scholarship (2020–2023); Best Undergraduate Thesis Award; Outstanding Graduate, NWPU; Hackathon Winner — NWPU AI Challenge.

MD RAKIBUL ISLAM RAIHAN (杨瀚)

AI 工程师 | 大模型推理与 GPU 系统 | 多模态 AI 研究 — 2027届应届 (中国 / 香港)

Email: raihanrakib.143@gmail.com | Phone: +86-17722362949 | Xi'an, China

LinkedIn: linkedin.com/in/theraihanrakib | GitHub: github.com/theraihanrakibb

Portfolio: theraihanrakib.github.io/Online-Portfolio/

Languages: English (Fluent) | Chinese (Conversational) | Bangladesh

个人简介

西北工业大学软件工程硕士在读 (前 1%)，具备大模型推理系统、GPU 优化与多模态 AI 研究的工程实战。IEEE TNNLS (Q1) 论文第一作者 (NVFP4-DiT, 审稿中) —— **4 比特音频引导视频扩散 Transformer**，低成本 AIGC 视频生成 (显存降 4 倍)。曾任 Infix.ai AI 基础设施工程师实习：在 4× A800 集群构建生产级大模型服务——三级 KV-cache 与 RadixAttention 前缀缓存 (**92.27% 命中率、8.3倍 TTFT 降低**)、基于 Mooncake RDMA 的 PD 分离服务 (**192→256 并发**)、FP8 量化的 CUDA/Triton 算子优化、RDMA 上 NCCL 分析。研究涵盖视频/图像生成、多模态深度伪造检测与低精度 (FP8/FP4/NVFP4) 推理。Python/C++/CUDA; PyTorch; SGLang; Linux。

工作经历

AI 基础设施工程师实习 | 无界智索 (深圳) 科技 | 深圳 | 2026年4月 – 2026年6月

- 修改 SGLang 源码 (缓存管理与内存分配模块: kvcacheio.py、hiradix_cache.py、memory_pool_host.py)，解决生产缓存故障，并在 4× A800 集群支持混合注意力 Qwen 模型部署。
- 设计三级 KV-cache 架构 (GPU HBM → CPU DRAM → SSD)，基于 SGLang 与 Mooncake RDMA，实现 **92.27% 命中率**，TTFT 降低**8.3倍 (55秒 → 6.7秒)**。
- 使用 Nsight Systems 分析 CUDA/Triton 推理流水线，定位 GEMM 瓶颈，通过算子优化与**FP8 量化**提升服务效率。
- 构建 PD 分离服务架构 (RDMA 共享 KV)，并发稳定性从 **192 提升至 256** 请求。
- 排查并解决**12+**起生产问题 (RDMA、etcd、网络、存储、分布式推理系统)。

软件工程师实习 | 香港阿姆兰集团 | 深圳 | 2025年6月 – 2025年9月

- 基于 React、REST API 与后端自动化服务开发生产级全栈应用。
- 开发 AI Agent 工作流自动化系统，运营效率提升 **25%**。

电气软件工程师实习 | 陕西龙工智能科技 | 西安 | 2025年2月 – 2025年4月

- 基于 CODESYS 开发 PLC 工业控制系统，用于机器人与工业自动化。
- 实现 Modbus、CAN 与 Ethernet/IP 通信协议，电机控制性能提升 **15%**。

研究经历

研究生研究员 | 西北工业大学 | 2024年9月 – 至今

- 开发**NVFP4-DiT: 4 比特音频引导视频扩散 Transformer**，显存降低 4 倍且保持生成质量。
- 实现多模态扩散 Transformer 的**低精度量化 (FP8/FP4/NVFP4)**，评估质量-效率权衡。
- 第一作者，《NVFP4-DiT: 面向低成本视频生成的高效 4 比特音频引导视频扩散 Transformer》(IEEE TNNLS, 审稿中)。
- 在 **A800/H100** 上对 NVFP4-DiT 进行基准测试，使用 PyTorch Profiler 与 Nsight 分析端到端延迟与显存，指导低精度算子选型。
- 基于视觉、音频与时序一致性的多模态深度伪造检测研究。
- 硕士论文：《检测深度伪造视频：一种利用视觉、听觉与时序不一致性的多模态 AI 框架》。

本科研究员 | 西北工业大学 | 2023年1月 – 2024年7月

- 基于 Apache Spark 与 CKKS 同态加密设计分布式机密查询框架。
- 通过优化分析管线实现**95% 隐私保护与20% 性能提升**。
- 本科论文：《面向 Apache Spark 的分布式机密查询协议设计与实现》。

技术技能

编程语言: Python、C++、Java、SQL、Bash、CUDA C++

大模型推理与服务: SGLang (RadixAttention、分页注意力、连续批处理)、vLLM、TensorRT-LLM、KV Cache 设计、PD 分离、前缀缓存、Mooncake RDMA、NCCL、etcd、NATS

GPU 与性能优化: CUDA、CUTLASS、Triton、FlashAttention-2/3、FP8/FP4/INT4 量化、GEMM 优化、Nsight Systems/Compute、PyTorch Profiler

ML 与分布式训练: PyTorch、DeepSpeed、Megatron-LM、LoRA/QLoRA、ZeRO、分布式训练/推理、Apache Spark

生成式视觉与多模态: 扩散 (DDPM、DiT)、音频引导视频生成、AIGC 视频/图像、多模态理解与融合、深度伪造检测、OpenCV
基础设施 / 熟悉: Linux、Docker、Kubernetes、Slurm、Prometheus、Grafana、TCP/IP、Git、CI/CD、AWS/阿里云、ML 编译器 (XLA/TVM)、TensorRT

教育背景

软件工程硕士，西北工业大学 (985/211) — 2024年9月 – 2027年3月

软件学院 · GPA **88/100** · 前 1% · 中国政府奖学金

计算机科学与技术学士，西北工业大学 (985/211) — 2020年9月 – 2024年7月

计算机学院 · GPA **85/100** · 前 1% · 校长奖学金 · 最佳本科毕业论文奖

论文与荣誉

论文：第一作者，《NVFP4-DiT: 面向低成本视频生成的高效 4 比特音频引导视频扩散 Transformer》— IEEE TNNLS (Q1)，审稿中。

荣誉：中国政府奖学金 (2024–2026)；校长奖学金 (2020–2023)；最佳本科毕业论文奖；西北工业大学优秀毕业生；NWPU AI 挑战赛黑客松冠军。